

Module 1: Introduction to TypeScript

1. What is TypeScript?

TypeScript is a superset of JavaScript that adds static types. It provides better tooling, improves maintainability, and helps catch errors during development.

Advantages of TypeScript:

- Static typing helps prevent runtime errors.
- Supports modern JavaScript features.
- Provides better IDE support.
- Helps in large-scale application development.

2. Differences Between JavaScript and TypeScript

Feature	JavaScript	TypeScript
Typing	Dynamic	Static
Compilation	Interpreted	Compiled
Error Detection	Runtime	Compile-time
Object-Oriented	Limited	Strong Support

3. Installing TypeScript & Setting Up the Environment

To install TypeScript globally, use:

```
npm install -g typescript
```

Check installation:

```
tsc --version
```

4. First TypeScript Program

Create a file `hello.ts`:

```
typescript
```

```
let message: string = "Hello, TypeScript!";  
console.log(message);
```

Compile it using:

```
tsc hello.ts
```

Run:

```
node hello.js
```

5. Compiling TypeScript Code

TypeScript files (`.ts`) must be compiled into JavaScript (`.js`) using the TypeScript compiler (tsc).

Command:

```
tsc filename.ts
```

6. Understanding TypeScript Configuration (tsconfig.json)

A `tsconfig.json` file helps configure the TypeScript compiler.

Example:

```
json
{
  "compilerOptions": {
    "target": "ES6",
    "outDir": "./dist",
    "strict": true
  },
  "include": ["src"]
}
```

Practice Questions:

1. What is the main advantage of using TypeScript over JavaScript?
2. How do you compile a TypeScript file?
3. What command installs TypeScript globally?

4. What is the role of `tsconfig.json`?

5. Convert the following JavaScript code to TypeScript:

```
javascript
```

```
function greet(name) {  
  return "Hello, " + name;  
}  
console.log(greet("Alice"));
```

Practice Answers:

1. TypeScript provides static typing, which helps prevent runtime errors.

2. Use the `tsc filename.ts` command.

3. `npm install -g typescript`.

4. It configures the TypeScript compiler settings.

5. Converted TypeScript code:

```
typescript
```

```
function greet(name: string): string {  
  return "Hello, " + name;  
}  
console.log(greet("Alice"));
```